

Name _____ Date _____ Period _____

Comparing All Seven Parent Functions - Practice

Algebra II | Unit 1 | Lesson 8 Practice | TEK A2.2.A

Objective: I can compare the seven parent functions, name one from its attributes, and find relative max/min on an interval.

Section 1 — Recall

1. Which parent function has no intercepts at all? Which two have a minimum?

No intercepts: _____ Have a minimum: _____

2. True or false: every parent function with an asymptote also has domain all real numbers.

Section 2 — Apply

3. A function has domain $(0, \infty)$, range all reals, and an x-intercept. Name it.

4. Two parent functions share the same domain, range, and symmetry (both all reals, rotational symmetry). Name both, and give one way to tell them apart using a single point.

Names: _____ Point: _____

Section 3 — Challenge

5. $f(x) = |x|$ on the interval $[-4, 2]$. Give the minimum and maximum, explaining why the maximum is not simply "whichever endpoint has the larger x-value."

min: _____ max: _____ at x = _____

6. A graph has an asymptote, two separate branches, no intercepts, and is symmetric about the origin (rotational). Which single parent function fits all these clues at once?

Section 4 — Explain It

7. A classmate says: "If a function has no maximum overall, it can never have a relative maximum on any interval either." Use x^3 on a closed interval to test this claim.

